

Subject Description Form

Subject Code	EIE2211
Subject Title	Logic Design
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To provide students with a broad view in both hardware and software aspects of digital systems in general and microprocessor systems in particular, and enable them to gain understanding and skills that will be used in later computer related courses. Emphasis will be placed on topics including: 1. Common binary logic components found in a microcomputer system 2. Use and applications of programmable logic devices 3. Structure and organization of microprocessors
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the fundamentals of digital systems and associated technologies. 2. Analyse and design simple systems related to digital logic. 3. Apply logic design techniques to construct digital systems with programmable logic devices and microprocessors, and appreciate the use of them. 4. Appreciate the importance of creativity and critical thinking on finding “good” solutions or making “good” designs. <u>Category B: Attributes for all-roundedness</u> 5. Think critically.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Logic Circuit and ICs</u> 1.1 Decoders and encoders 1.2 Multiplexers and demultiplexers 1.3 Binary adders, binary adder-subtractors 1.4 Binary multipliers 1.5 Sequential circuit analysis and design 1.6 Registers and counters 1.7 HDL representation. 2. <u>Memory and Programmable Logic Devices</u> 2.1 RAM: Write and read operations, timing waveforms, RAM integrated circuits, three-state buffers, DRAM ICs 2.2 Programmable logic technologies 2.3 ROM, PLA and PAL 2.4 VLSI programmable logic devices: Xilinx FPGA.

	3. <u>Microprocessor</u> 3.1 Register transfer operations 3.2 Microoperations 3.3 Bus-based transfer 3.4 ALU 3.5 Shifter 3.6 Datapath representation 3.7 Control word 3.8 Control unit 3.9 Hardwired control 3.10 Basic Assembly Language Programming. Laboratory Experiment: 1. Basic logic gates and their applications 2. Hardware description language and programmable logic devices																																																											
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	The continuous assessment will consist of several quizzes, 1 lab demonstration, 1 lab report, and 1 test. Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: <table><tr><th>Specific Assessment Methods/Tasks</th><th>Remark</th></tr><tr><td>Quizzes, tests and examination</td><td>End-of chapter type problems used to evaluate students’ ability in applying concepts and skills learned in the lessons. Students need to think critically and creatively in order to come up with solutions for existing problems.</td></tr><tr><td>Laboratory sessions</td><td>Each student is required to do a demonstration and submit a lab report.</td></tr></table>		Specific Assessment Methods/Tasks	Remark	Quizzes, tests and examination	End-of chapter type problems used to evaluate students’ ability in applying concepts and skills learned in the lessons. Students need to think critically and creatively in order to come up with solutions for existing problems.	Laboratory sessions	Each student is required to do a demonstration and submit a lab report.
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Student Study Effort Expected	Class contact (time-tabled):							
	• Lecture	24 Hours						
	• Tutorial/Laboratory/Practice Classes	15 hours						
	Other student study effort:							
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination	36 Hours						
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	30 Hours						
	Total student study effort:	105 Hours						
Reading List and References	Textbooks: 1. M.M. Mano and C.R. Kime, <i>Logic and Computer Design Fundamentals</i> , 4 th ed., Upper Saddle River, NJ: Prentice-Hall, 2008.							
	Reference Books: 1. M.M. Mano and M.D. Ciletti, <i>Digital Design</i> . Upper Saddle River, NJ: Prentice-Hall, 2007. 2. S. Yalamanchili, <i>VHDL – A Starter’s Guide</i> , 2 nd ed. Prentice-Hall, 2005. 3. E.O. Hwang, <i>Digital Logic and Microprocessor Design With VHDL</i> , 1 st ed., CL-Engineering, 2006.							